CHEM 403: GENERAL CHEMISTRY
Fall 2020

Instructor: Dr. Samantha Reynolds
Email: Samantha.Reynolds@unh.edu
Office: n/a – working remotely
Office Hours: By appointment - please feel free to contact me by email with questions or to set up a meeting time on Zoom.

Lecture: Mondays 1-4pm
Lab, Three In-person Sections: Tuesdays 6-9pm; Wednesdays 9am-12pm; Thursdays 6-9pm
(Lab Instructors: John Welsh and Gordon Hunter)
One Online Section: Hours Arranged (Lab Instructor: Sarah Prescott)

Required Materials:

Lecture Text – Chemistry: Atoms First. OpenStax. ISBN-13: 978-1-938168-15-4.

This is an open access textbook, available online. You may access chapters directly online, download a pdf of the text for free, or order a hard copy. We will be using this text for both semesters. You will also need a scientific calculator for use in class and on tests and exams.

Course Objectives:

The objective for this course is to teach the fundamental concepts and physical laws of chemistry. At the end of this first semester, a student will be familiar with the components of matter, including atoms, molecules, mixtures and solutions. A student will be able to discuss periodic trends, depict chemical bonding, write and balance reactions, calculate stoichiometry, and perform calculations involving thermochemical reactions and gas laws. A student will become proficient in using SI units and showing work for calculations.

Grades:

LECTURE:

The lecture grade is based on a midterm and final exam, worth 200 points each, weekly online homework sets worth 20 points each, and team assignments, called Team Builds worth 50 points each. While the midterm and final exam will primarily focus on new material covered in that section, please recognize that chemistry is a highly cumulative subject, and many principles and topics throughout the entire year will build on previously learned material. The lecture is worth 80% of your overall CHEM 403 grade.

Lecture Points Overview	
Weekly Homework	
10 x 20 points	200 points
Team Builds	
4 x 50 points	200 points
Midterm	200 points
Final Exam	200 points
TOTAL	800 points

The best way to understand the concepts in this course is to practice them. Homework sets will be assigned most weeks to be completed online before the next lecture meeting. You are welcome to freely collaborate on these homework sets with your peers, course tutors, or by using other resources. Team Builds will be assigned as out-of-class assignments. A portion of these assignment grades is based on participation.

Additional non-graded assignments in the lecture include reading assigned chapters, watching pre-lecture videos, and working practice problems in preparation for each class meeting. I expect that you are coming to each lecture having familiarized yourself with the course material and having begun to work on that lecture's problem set.

An extra credit assignment will be available that will be worth 25 points. Details of the assignment will be announced mid-semester.

LABORATORY:
The CHEM 403 General Chemistry laboratory is a supplement to the lecture. As such, it counts as 20% of your total grade in the course. Please note that you must pass the lab in order to pass the course. A syllabus for the laboratory will be distributed separately by your lab instructor.

ASSIGNMENT OF A LETTER GRADE:
After your lecture and lab grades have been combined, your overall letter grade for the course will be based on the percentage grading structure below.

A = 93.0-100	B- = 80.0-82.9	D+ = 65.0-69.9
A- = 90.0-92.9	C+ = 77.0-79.9	D = 60.0-64.9
B+ = 87.0-89.9	C = 73.0-76.9	F = 59.9 and below
B = 83.0-86.9	C- = 70.0-72.9	

Late Work:

It is imperative that you keep up with assigned homework each week. Homework will be available in the style of an online "quiz" that will be available to submit until the next lecture meeting. At that time, no late submissions will be accepted, and students that have not submitted answers to the homework set will receive a zero grade for that assignment. Team Build submissions will be subject to a 5 point (10%) deduction for every half day (defined as a 12-hour period from the due date and time) the assignment is late.

Attendance:

Missing even one lecture may put a student at a significant disadvantage in the course. Please plan to attend all online lectures. If you are unable to attend a lecture, recordings of the Zoom sessions will be made available following each session; it is your responsibility to review this lecture content and come to me with any follow-up questions on the material. Please note that I cannot reteach a full lecture during office hours.

If circumstances beyond a student's control prevent him or her from completing a homework assignment, working on a team build, or taking an exam as scheduled, special arrangements can be discussed so that the student is not penalized for an excused absence.

Please let me know as soon as possible if you will have to miss an assignment or exam due date due to a legitimate excuse. If circumstances do not allow you to make prior arrangements for your absence, please contact me as soon after the missed assignment as you are able.

myCourses (Canvas):

myCourses is UNH's course management system. You can access your UNH myCourses page at mycourses.unh.edu and use your UNH ID and password to log in. All course announcements, assignments, and documents, including homework "quizzes", lecture problem sets, and answer keys will be made available on myCourses during the semester. Academic Computing Services can help if you have issues with access to myCourses; it is your responsibility to resolve any technical issues.

Zoom Policies:

I would like to begin Zoom sessions by seeing everyone. If you are not in a situation where you can join with a webcam, please set your profile picture to a current (school-appropriate) photo. After I switch to sharing my screen and begin to cover lecture content, you are welcome to turn off your webcam.

During the Zoom session, be prepared to engage with me and your classmates. This may include being called on or asked to use the "raise hand" feature. You can have your audio muted as a default, especially if there is background noise where your computer is set up, but you should be ready to turn it on to answer questions, or to ask any questions you have for me.

I will announce when I begin recording the Zoom session and when I am no longer recording. Your enrollment in a UNH course is consent to being recorded by UNH media platforms for educational and remote access purposes. The University and Zoom have FERPA-compliant agreements in place to protect the security and privacy of UNH Zoom accounts. Students may not share recordings outside of their course.

Academic Honesty:

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (http://www.unh.edu/vpsas/handbook/academic-honesty). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work – in whole or in part – as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.
Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes, the textbook, or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted.

For online exams, you will be given instructions about which materials are and are not permitted to be used. You will be asked to digitally sign an Academic Honesty statement before beginning any exam agreeing to follow all exam policies.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Academic Support/Tutoring:

The Center for Academic Enrichment is a resource for all students enrolled in UNH Manchester courses. Through a collaborative tutoring approach, they aim to provide students with opportunities to maximize their learning potential, creating life-long learners for academic and professional success. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in math and science courses. ***All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course.*** In addition, peer and professional tutors may run study groups for you class and provide drop-in math tutoring. Please visit the CAE online (http://manchester.unh.edu/current-students/cae-more-powerful-learning) to learn more about their services or to make an appointment with a tutor.

Disability Services at UNH Manchester:

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Student Accessibility Services (SAS) office. The Student Accessibility Coordinator at UNHM is Jenessa Zurek. Please reach out to the SAS office via email at jenessa.zurek@unh.edu for registration information and disability related questions. Jenessa Zurek is available through phone and email Mondays and Wednesdays from 9am-2pm.

Mental Health Services at UNH Manchester:

In partnership with The Mental Health Center of Greater Manchester, UNH Manchester offers free mental health sessions for students. Students can schedule virtual counseling sessions by emailing unhm.advising@unh.edu. Counselors will be available virtually on Monday, Tuesday, and Thursday from 9am-5pm.

Class Cancellations and Delayed Openings:

UNH Manchester makes weather-related decisions independent of UNH Durham. For information on cancellations and closings, call the UNH Manchester Information Line at **603 641 4100**, or visit the UNH Manchester website. Closing and delay information will also be announced on the following stations:
WMUR Channel 9 TV; WGIR 101 FM and 610 AM; WJYY 105.5 FM; WMLL 96.5 FM; WOKQ 97.5 FM; WZID 95.7 FM; WFEA 1370 AM.

While this is an online course, I understand some of you may need access to the Manchester campus for technology resources. If campus is closed due to weather, I will also cancel our synchronous lecture meeting.

TENTATIVE LECTURE SCHEDULE
CHEM 403 - Fall 2020

Date	Lecture Topics	Textbook Chapters	Assignments
8/31	Course Introduction, Phases of Matter, Measurements	1.1-1.4, 1.6	
9/7	*Labor Day – No Class*		
9/14	Significant Figures, Atomic Theory	1.5, 2.1-2.3	HW 1 Due
9/21	Electronic Structure	[3.1-3.2], 3.3-3.4	HW 2, Team Build I Due
9/28	Periodic Properties, Ionic Bonding	3.5-3.7, 4.1, 4.3	HW 3 Due
10/5	Covalent Bonding, Molecular Geometry	4.2-4.6	HW 4 Due
10/12	Composition of Substances	2.4, 6.1-6.2	HW 5, Team Build II Due
10/19	Stoichiometry of Reactions	7.1, 7.3-7.4	HW 6 Due
10/26	**Midterm**	**Chapters 1-4, 6-7**	
11/2	Solution Chemistry & Molarity	6.3	
11/9	Precipitation Reactions	7.1-7.2	HW 7 Due
11/16	Thermochemistry	9.1-9.2	Team Build III Due
11/23	Thermochemistry (cont.)	9.3	HW 8 Due
11/30	Gas Laws	8.1-8.2	HW 9, Team Build IV Due
12/7	Gas Laws (cont.)	8.3, 8.5-8.6	HW 10, Bonus Due
12/14	*University Reading Day (no class)*		
12/21	**Final Exam**	**Chapters 6-9**	